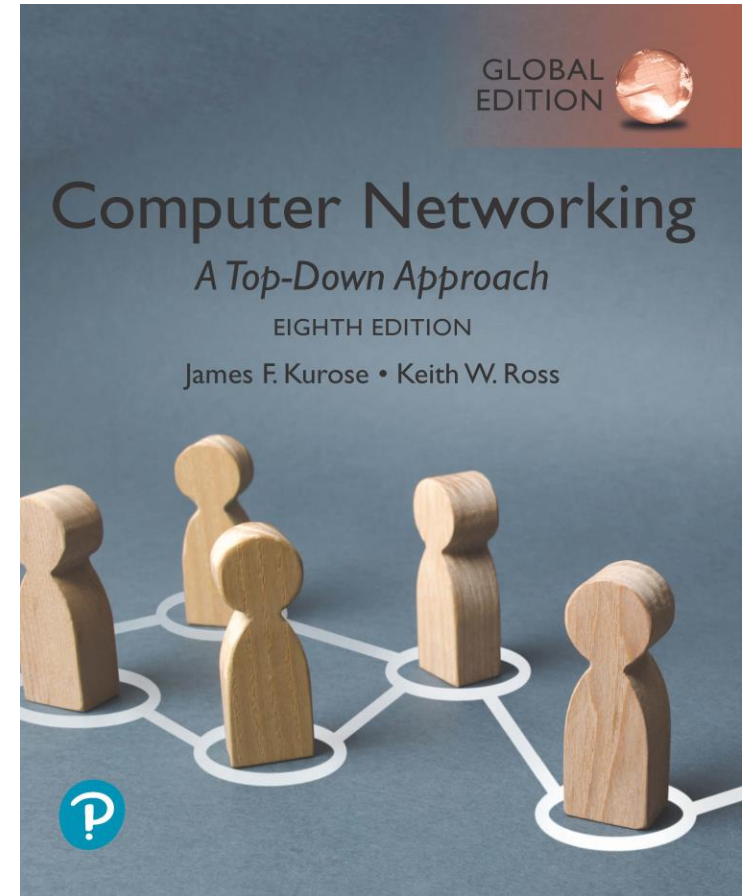


Chapter 8

Security



Computer Networking: A Top-Down Approach

8th edition, Global Edition

Jim Kurose, Keith Ross

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Chapter 8 outline

- What is network security?
- Principles of cryptography
- Authentication, message integrity
- Securing e-mail
- Securing TCP connections: TLS
- **Network layer security: IPsec and VPN**
- Security in wireless and mobile networks
- Operational security: firewalls and IDS

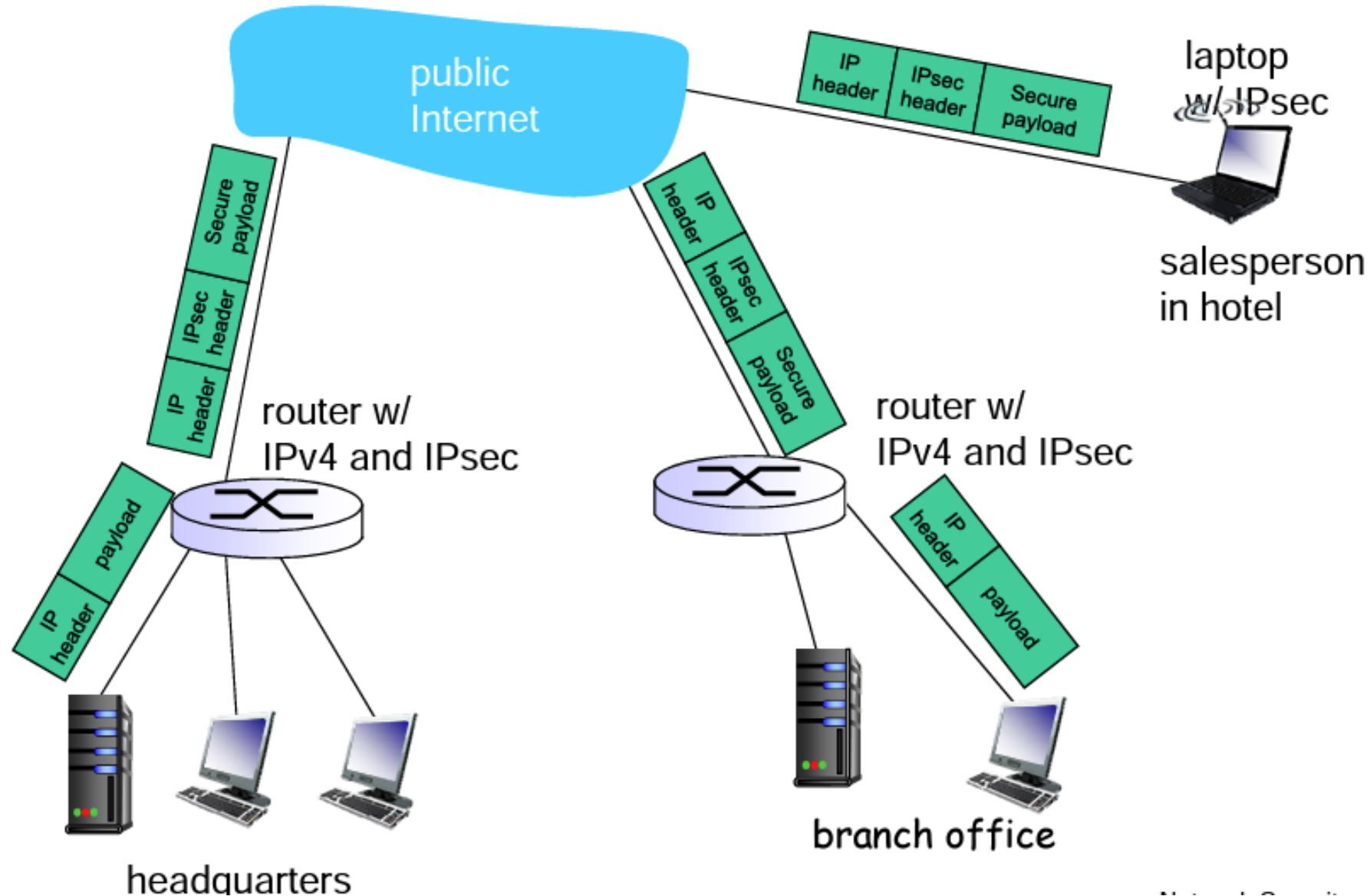


Virtual Private Networks (VPNs)

motivation:

- ❖ institutions often want private networks for security.
 - costly: separate routers, links, DNS infrastructure.
- ❖ VPN: institution's inter-office traffic is sent over public Internet instead
 - encrypted before entering public Internet
 - logically separate from other traffic

Virtual Private Networks (VPNs)

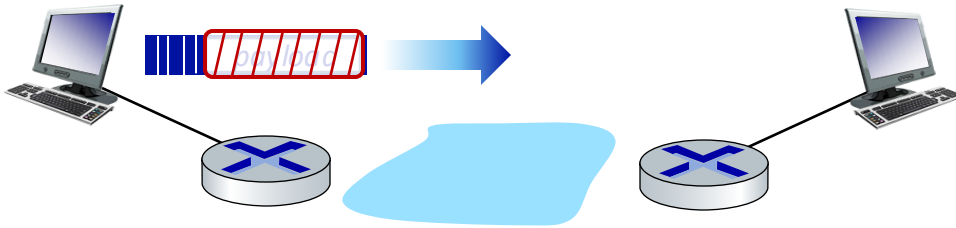


IPsec services

- ❖ data integrity
 - ❖ origin authentication
 - ❖ replay attack prevention
 - ❖ confidentiality
-
- ❖ two protocols providing different service models:
 - AH
 - ESP

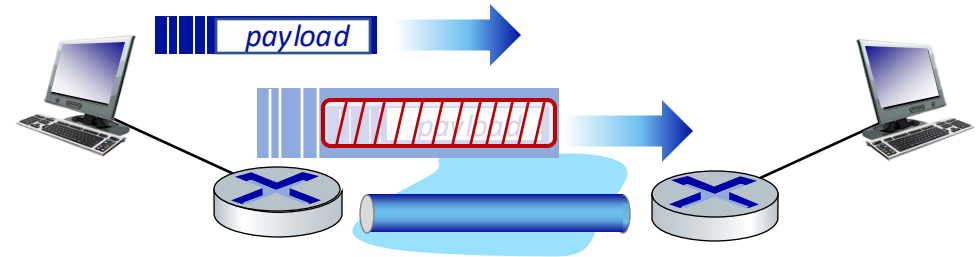
IP Sec

- provides datagram-level encryption, authentication, integrity
 - for both user traffic and control traffic (e.g., BGP, DNS messages)
- two “modes”:



transport mode:

- *only* datagram *payload* is encrypted, authenticated



tunnel mode:

- entire datagram is encrypted, authenticated
- encrypted datagram encapsulated in new datagram with new IP header, tunneled to destination

Two IPsec protocols

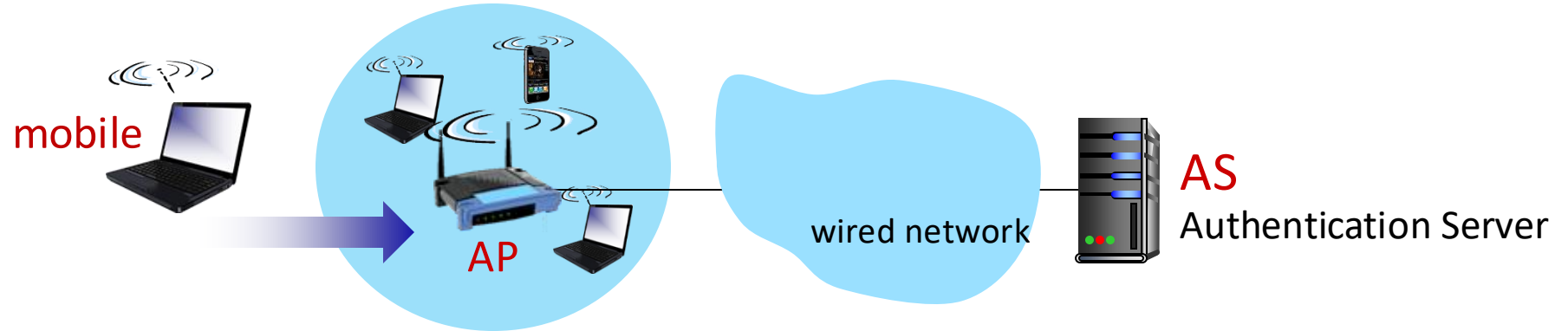
- Authentication Header (AH) protocol [RFC 4302]
 - provides source authentication & data integrity but *not confidentiality*
- Encapsulation Security Protocol (ESP) [RFC 4303]
 - provides source authentication, data integrity, and *confidentiality*
 - more widely used than AH

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- **Security in wireless and mobile networks**
 - 802.11 (WiFi)
 - 4G/5G
- Operational security: firewalls and IDS



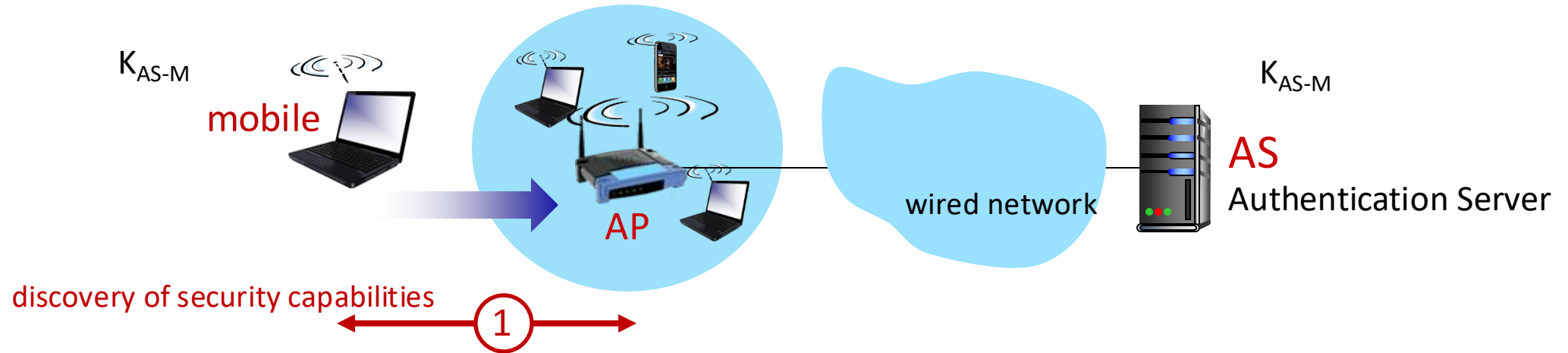
802.11: authentication, encryption



Arriving mobile must:

- **associate** with access point: (establish) communication over wireless link
- **authenticate** to network

802.11: authentication, encryption

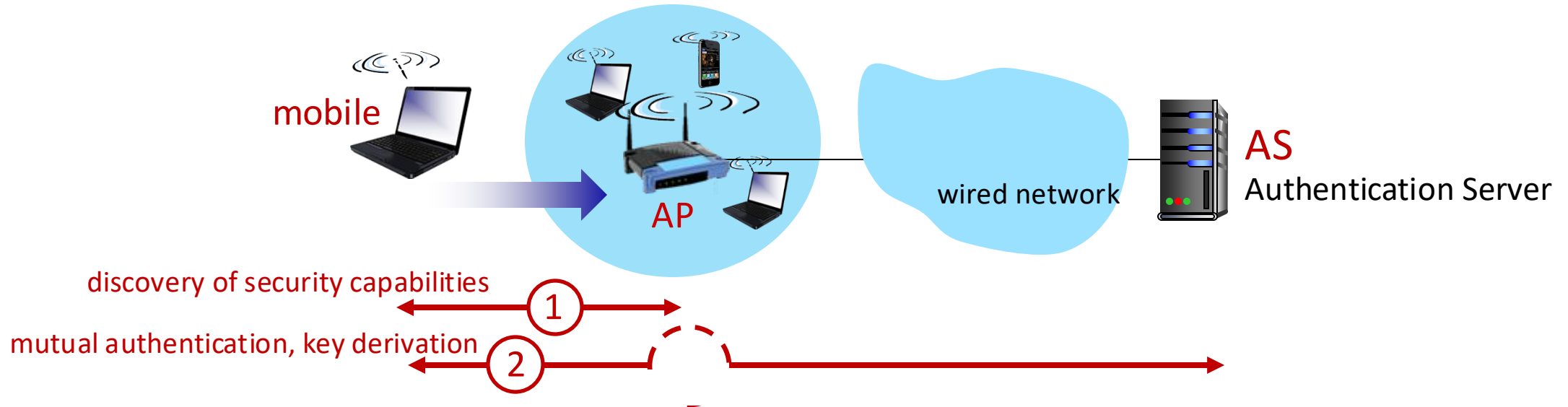


① discovery of security capabilities:

- AP advertises its presence, forms of authentication and encryption provided
- device then requests the specific form of authentication, encryption desired

although device and AP already exchanging messages, device not yet authenticated, does not have encryption keys (med unntak av et passord K_{AS-M})

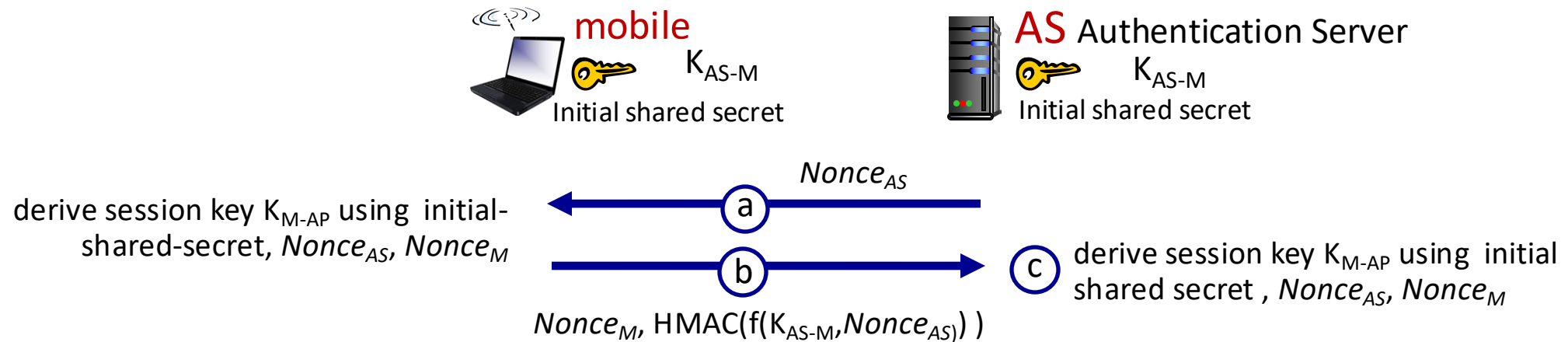
802.11: authentication, encryption



② mutual authentication and shared symmetric key derivation (ex WPA3):

- AS and mobile already have shared common secret (e.g., password K_{AS-M})
- AS and mobile use shared secret, nonces (number used once, to prevent relay attacks), cryptographic hashing (ensure message integrity) in authenticating each other
- AS and mobile derive symmetric session key to be used for frame encryption

802.11: WPA3 handshake



2

(a) AS generates $Nonce_{AS}$, sends to mobile

(b) mobile receives $Nonce_{AS}$

- generates $Nonce_M$
- generates symmetric shared session key K_{M-AP} using $Nonce_{AS}$, $Nonce_M$, and initial shared secret, its MAC address and MAC address of AS
- sends $Nonce_M$, and HMAC-signed value using $Nonce_{AS}$ and initial shared secret

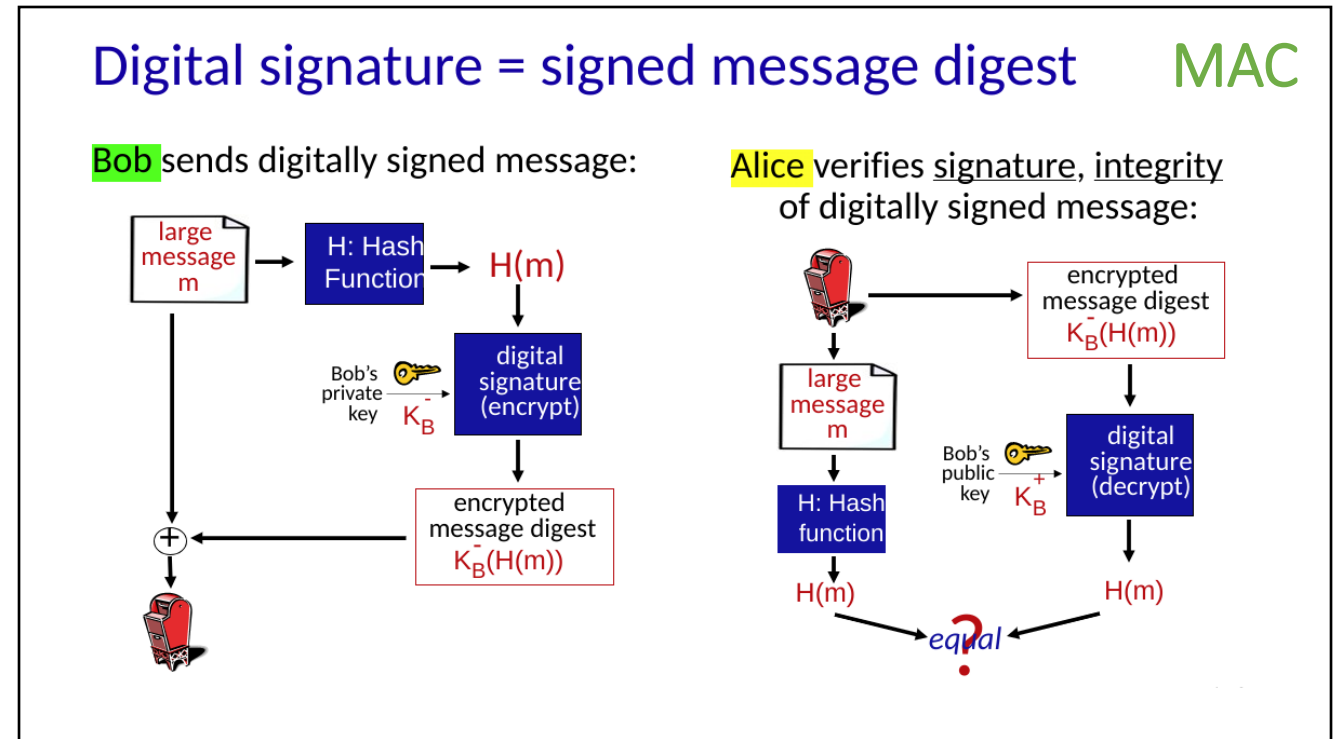
(c) AS derives symmetric shared session key K_{M-AP}

Vent nå litt.....

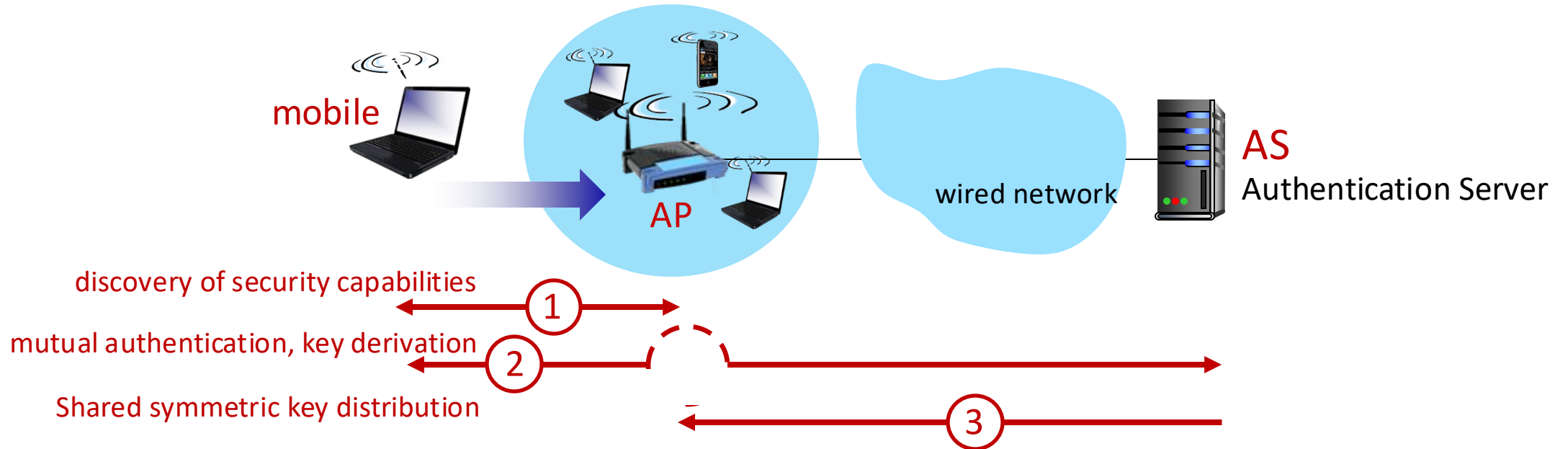
HMAC ? MAC ? ...Message Auth Code...hæh ?

Header Message Auth Code
– HMAC er en av
maaaaange MAC varianter

Jfr kap 8.3.2

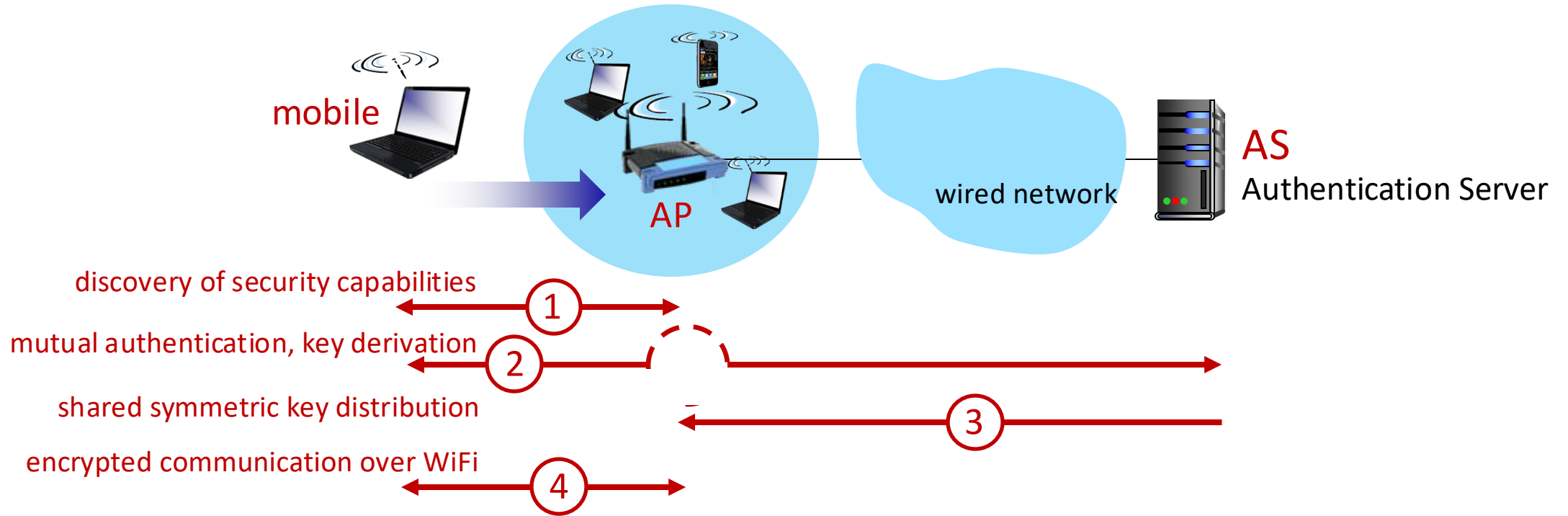


802.11: authentication, encryption



- ③ shared symmetric session key distribution (e.g., for AES encryption)
- AS informs AP of the shared symmetric session

802.11: authentication, encryption



④ encrypted communication between mobile and remote host via AP

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Authentication, encryption in 4G LTE



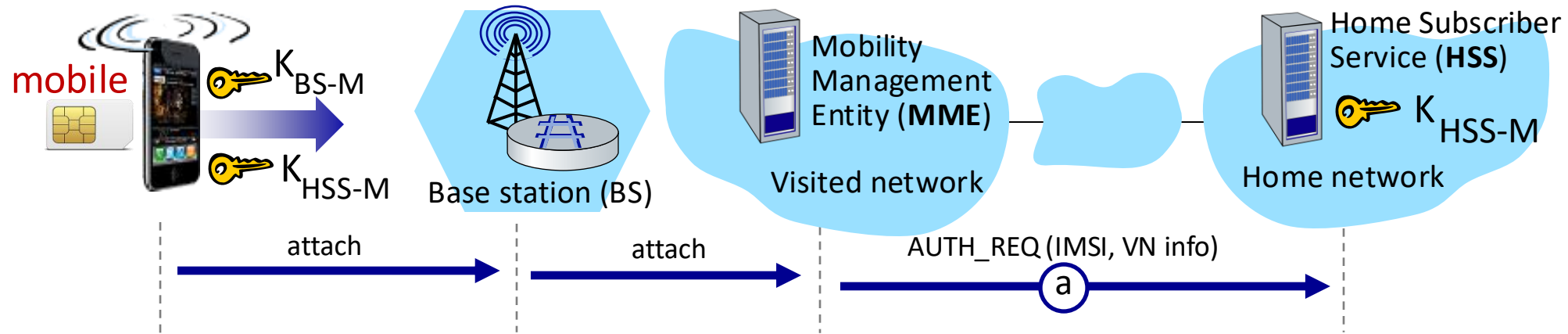
- arriving mobile must:
 - **associate** with BS: (establish) communication over 4G wireless link
 - **authenticate** itself to network, and **authenticate** network
- notable differences from WiFi
 - mobile's SIMcard provides global identity, contains shared keys
 - services in visited network depend on (paid) service subscription in home network

Authentication, encryption in 4G LTE



- mobile, BS use **derived session key K_{BS-M}** to encrypt communications over 4G link
- **MME in visited network + HSS in home network, together play role of WiFi AS**
 - ultimate authenticator is HSS
 - trust and business relationship between visited and home networks

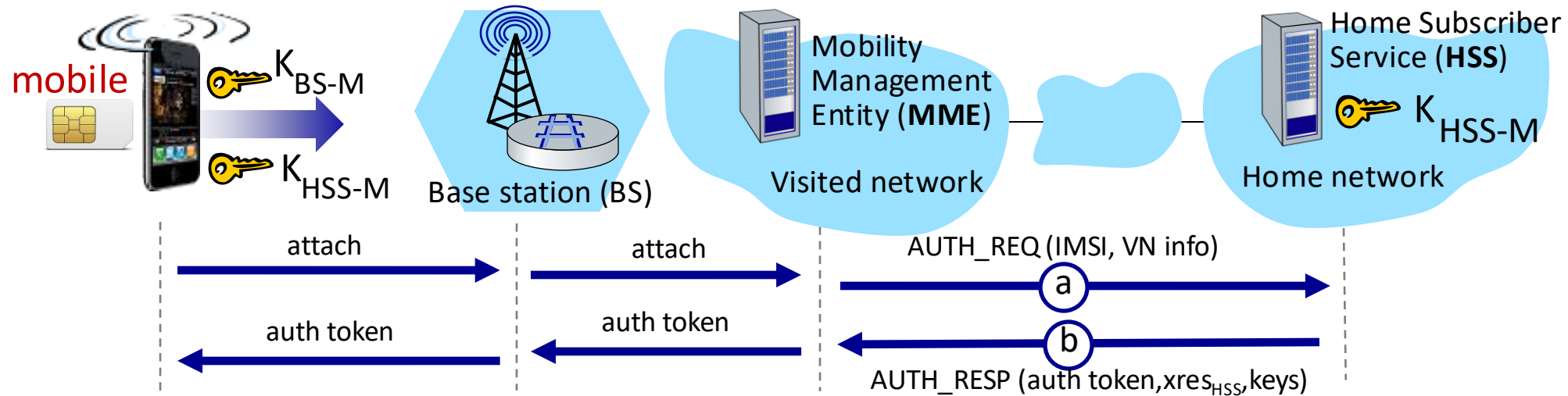
Authentication, encryption in 4G LTE



① authentication request to home network HSS

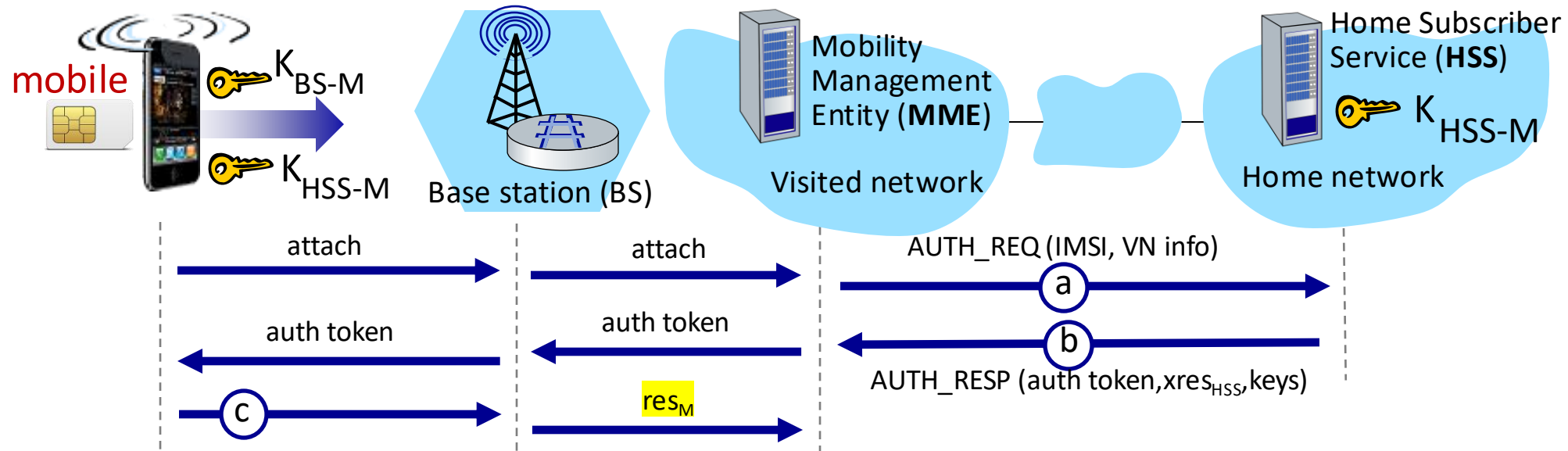
- mobile sends attach message (containing its IMSI, visited network VN info) relayed from BS to visited MME to home HSS
- **IMSI identifies mobile's home network**

Authentication, encryption in 4G LTE



- ② HSS use shared-in-advance secret key, K_{HSS-M} , to derive authentication token, *auth_token*, and expected authentication response token, $xres_{HSS}$
- *auth_token* contains info encrypted by HSS using K_{HSS-M} , allowing mobile to know that whoever computed *auth_token* knows shared-in-advance secret
 - mobile has authenticated network
 - visited HSS keeps $xres_{HSS}$ for later use

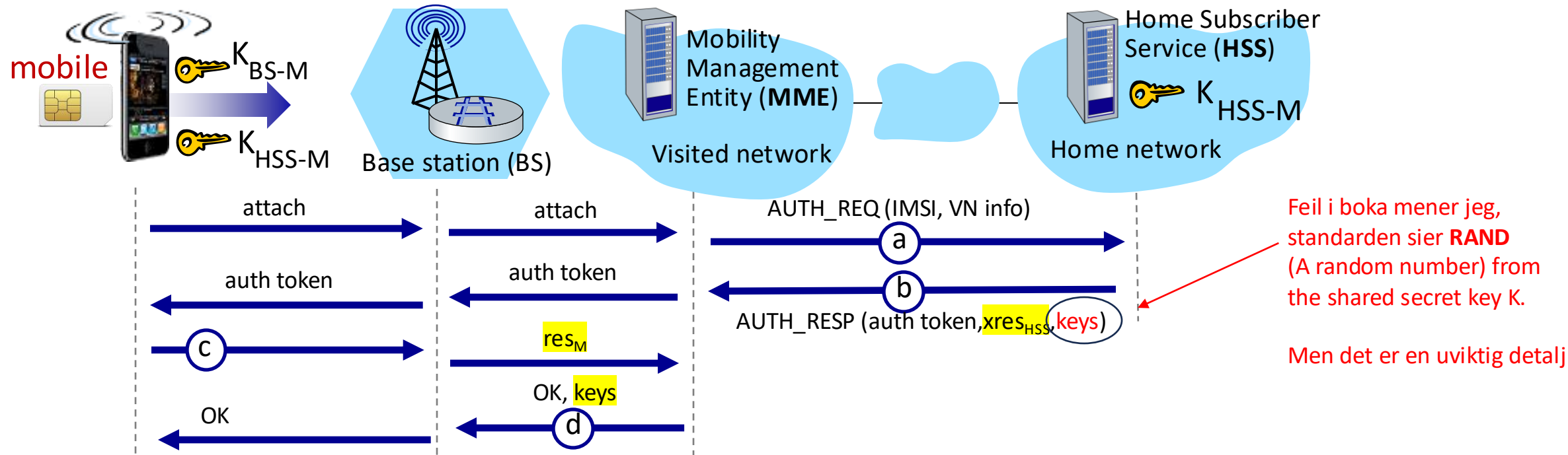
Authentication, encryption in 4G LTE



© authentication response from mobile:

- mobile computes res_M using its secret key to make same cryptographic calculation that HSS made to compute $xres_{HSS}$ and sends res_M to MME

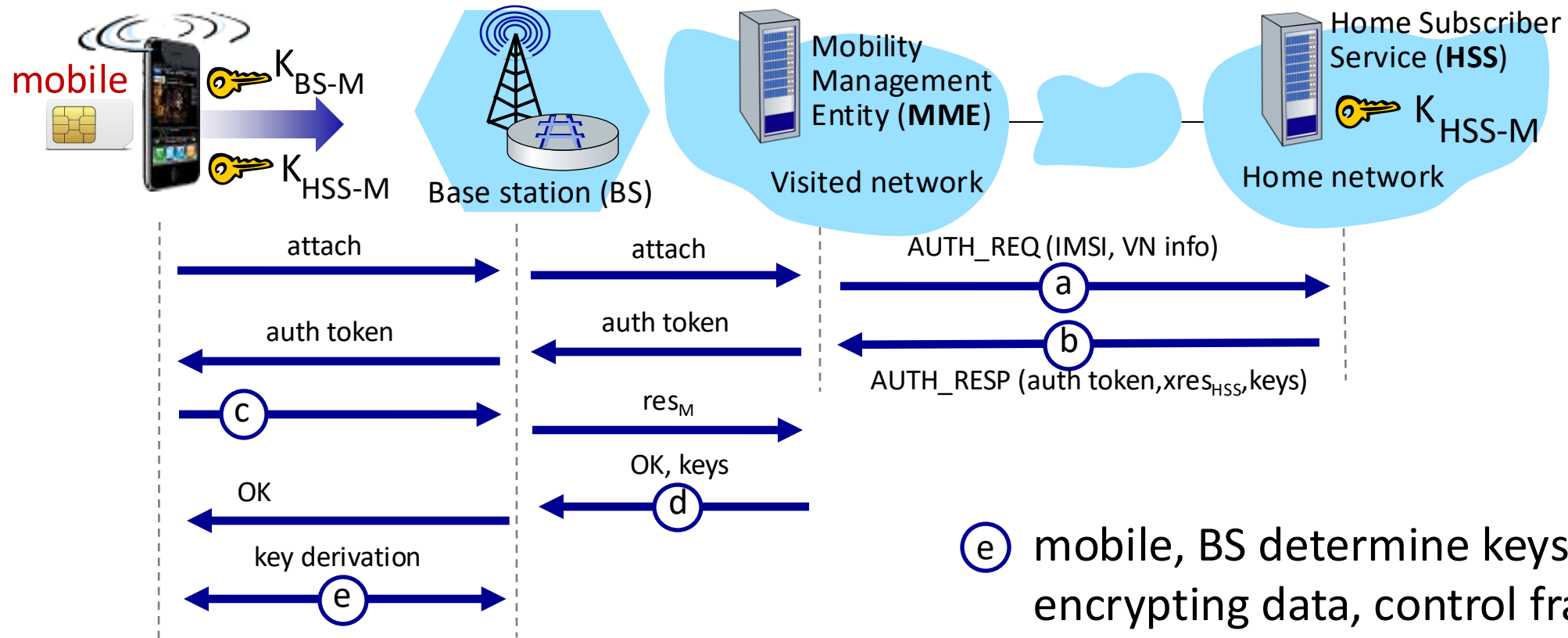
Authentication, encryption in 4G LTE



④ mobile is authenticated by network:

- MME compares mobile-computed value of res_M with the HSS-computed value of $xres_{HSS}$. If they match, mobile is authenticated !
- MME informs BS that mobile is authenticated, generates keys for BS

Authentication, encryption in 4G LTE



- ⑤ mobile, BS determine keys for encrypting data, control frames over 4G wireless channel
 - AES can be used

Authentication, encryption: from 4G to 5G

- **4G:** MME in visited network makes authentication decision
- **5G:** home network provides authentication decision
 - visited MME plays “middleman” role but can still reject
- **4G:** uses shared-in-advance keys
- **5G:** keys not shared in advance for IoT
- **4G:** device IMSI transmitted in cleartext to BS
- **5G:** public key crypto used to encrypt IMSI